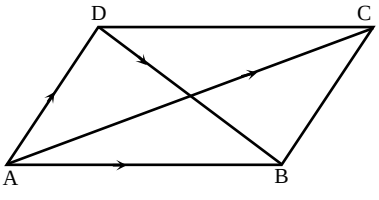
MATHEMATICS

SECTION – I

Single Correct Choice Type

1. (a) $\overrightarrow{DB} = \overrightarrow{DA} + \overrightarrow{AB}$ or $\overrightarrow{DA} = \overrightarrow{DB} - \overrightarrow{AB}$

$\therefore \quad \left(\overrightarrow{DA}\right)^2 = \left(\overrightarrow{DB}\right)^2 + \left(\overrightarrow{AB}\right)^2 - 2\overrightarrow{DB}.\overrightarrow{AB} \qquad \ldots(i)$



In parallelogram

$2\left(a^2 + b^2\right) = c^2 + DB^2 \Rightarrow \left(DB\right)^2 = 2a^2 + 2b^2 - c^2$

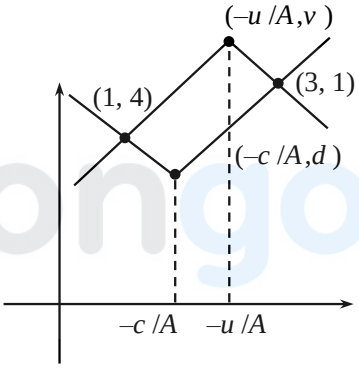
From (i) $\Rightarrow \quad b^2 = 2a^2 + 2b^2 - c^2 + a^2 - 2\overrightarrow{AB}.\overrightarrow{DB}$

$\overrightarrow{AB} \,.\, \overrightarrow{DB} = \dfrac{3a^2 + b^2 - c^2}{2}$

2. (b) The figure is a parallelogram and the diagonals bisect each other:

$\therefore \quad \left(\dfrac{-u}{A}\right) + \left(\dfrac{-c}{A}\right) = 3 + 1;$

$\therefore \quad \dfrac{u + c}{A} = -4$



3. (d) $f(x) = ax + b \qquad f'(x) = a > 0$

$\therefore \quad f(x)$ is an increasing function.

$\Rightarrow \quad f(-1) = 0$ and $f(1) = 2 \Rightarrow -a + b = 0;\ a + b = 2 \quad \Rightarrow \quad f(x) = x + 1$

Now $\cot\left(\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18\right)$

$= \cot\left(\tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{1}{8} + \tan^{-1}\dfrac{1}{18}\right) = \cot\left(\tan^{-1}\left(\dfrac{3}{11}\right) + \tan^{-1}\left(\dfrac{1}{18}\right)\right)$

$= \cot\left(\tan^{-1}\left(\dfrac{1}{3}\right)\right) = \cot\left(\cot^{-1}3\right) = 3 = f(2).$

4. (d) $I_2 = \displaystyle\int_1^{\operatorname{cosec}\theta} \frac{dx}{x\left(x^2+1\right)} = -\int_1^{\sin\theta} \frac{t}{1+t^2}dt$

$I_2 = -I_1$

$$\therefore \quad \begin{vmatrix} I_1 & I_1^2 & I_2 \\ e^{I_1+I_2} & I_2^2 & -1 \\ 1 & I_1^2+I_2^2 & -1 \end{vmatrix} = 0$$

5. **(a)** $[\sin x] = -1$ and $[\sqrt{2}\cos x] = -2$

or $-1 \le \sin x < 0$ and $-2 \le \sqrt{2}\cos x < -1$

$x \in (\pi, 2\pi)$ and $x \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right) \Rightarrow \quad x \in \left(\pi, \frac{5\pi}{4}\right).$

6. **(a)** $I = \lim\limits_{x\to\infty} \dfrac{\cot^{-1}\left(\dfrac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\dfrac{a^x}{\log_a x}\right)}$ as $\left(\dfrac{\log_a x}{x^a}\right) \to 0$ and $\left(\dfrac{a^x}{\log_a x}\right) \to \infty$

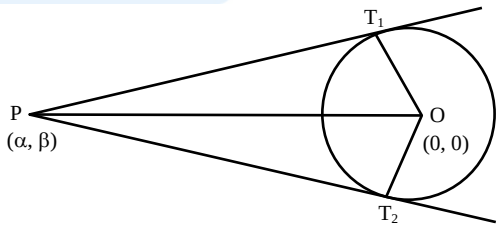
$\therefore \quad I = \dfrac{\pi}{2} \Big/ \dfrac{\pi}{2} = 1.$

7. **(d)** P, T_2, O, T_1 are concyclic points with PO as diameter

$\Rightarrow$ The circumcentre of ΔPT_1T_2 is $\left(\frac{\alpha}{2}, \frac{\beta}{2}\right)$

Now (α, β) lies on $px + qy + r = 0$

$\Rightarrow$ Locus of $\left(\frac{\alpha}{2}, \frac{\beta}{2}\right)$ is $2px + 2qy + r = 0$



8. **(c)** $t_{r+1} = \dfrac{x}{(rx+1)\{(r+1)x+1\}} = \dfrac{(r+1)x+1-(rx+1)}{(rx+1)[(r+1)x+1]} = \dfrac{1}{(rx+1)} - \dfrac{1}{(r+1)x+1}$

$$\therefore \quad S_n = \sum_{r=0}^{n-1} t_{r+1} = \begin{cases} 1 - \dfrac{1}{nx+1}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\therefore \quad f(x) = \lim_{n\to\infty} S_n = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0 \end{cases} \qquad \therefore \quad \lim_{x\to 0} f(x) = 1 \text{ and } f(0) = 0$$

Hence $f(x)$ is neither continuous nor differentiable at $x = 0$.

Clearly $f(x)$ is not a periodic function

SECTION – II
Multiple Correct Choice Type

9. (a), (b)

$$\arg\frac{z_1}{z_2}=\pm\pi \text{ and } |z_1+i(z_3-z_1)|=|z_1|+|z_3-z_1| \text{ iff } \arg\frac{z_1}{z_3-z_1}=\frac{\pi}{2}$$

So centre of circle = $\left(\frac{z_2+z_3}{2}\right)$ and ABC is right angle triangle.

10. (c), (d)

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$$

$da = 2R\cos A\, dA$, $db = 2R\cos B\, dB$,

$dc = 2R\cos C\, dC$

$$\frac{da}{\cos A}+\frac{db}{\cos B}+\frac{dc}{\cos C}=2R\,(dA+dB+dC)$$

also $A + B + C = \pi$

$\therefore$ $dA + dB + dC = 0$.

11. (c), (d)

$$\lim_{n\to\infty}\tan\left(\frac{1}{n}\right)\ln\left(\frac{1}{n}\right)=0 \qquad \text{[By L' Hospital's rule]}$$

$$\Rightarrow \quad f(x)=1$$

$$=\int\frac{dx}{\sin^{11/3}x\cos^{1/3}x}=\int\frac{\sec^4 x}{\tan^{11/3}x}dx=\int\frac{(1+\tan^2 x)\sec^2 x}{\tan^{11/3}x}dx$$

$$=\frac{-3}{8}(\tan x)^{-8/3}-\frac{3}{2}(\tan x)^{-2/3}+C$$

$$g\left(\frac{\pi}{4}\right)=-\frac{15}{8}.$$

12. (d)

Let O be the origin $\overrightarrow{OA}=\hat{i}+\hat{j}, \overrightarrow{OB}=\hat{i}-\hat{j}, \overrightarrow{OC}=p\hat{i}+q\hat{j}+r\hat{k}$

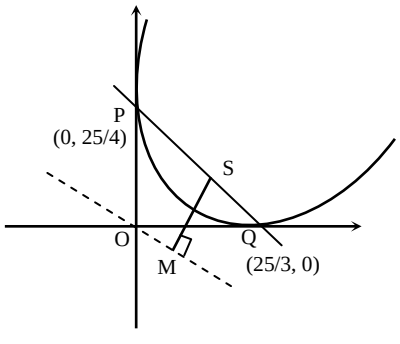
$\overrightarrow{AB}=-2\hat{j}, \overrightarrow{AC}=(p-1)\hat{i}+(q-1)\hat{j}+r\hat{k}$

A, B, C are collinear

$\therefore$ $\overrightarrow{AC}=\lambda\overrightarrow{AB}$.

SECTION–III
Comprehension Type

Sol.: Here OP and OQ are perpendicular tangents intersecting at origin we know that
(i) Point of intersection of perpendicular tangents lies on directrix of parabola (shown by doted lines and
(ii) Portion of tangents intercepted between the directrix and point of contact subtend right angle at focus.



Let focus is $S(h, k)$

So $\dfrac{k-\dfrac{25}{4}}{h}\times\dfrac{k}{h}=-1$ and $\dfrac{k}{h-\dfrac{25}{3}}\times\dfrac{k}{h}=-1 \Rightarrow h=3,\ k=4$

$PS=\dfrac{15}{4},\ QS=\dfrac{20}{3}$

If L = latus rectum

$\dfrac{4}{L}=\dfrac{1}{PS}+\dfrac{1}{QS} \Rightarrow \dfrac{4}{L}=\dfrac{4}{15}+\dfrac{3}{20} \Rightarrow L=\dfrac{240}{25}=\dfrac{48}{5}$

Circle passing through P, Q and O is $\left(x-\dfrac{25}{3}\right).x+y.\left(y-\dfrac{25}{4}\right)=0$

Equation of directrix is tangent to above circle at (0, 0).
Equation of directrix is $4x+3y=0$.

13. **(c)**
14. **(c)**
15. **(a)**

SECTION–IV
Matrix–Match Type

1. **A-1, 4** Clearly $\lim\limits_{x\to\frac{\pi}{2}}\left[\dfrac{|\sin x|-|\cos x|+1}{3}\right]=\left[\dfrac{2}{3}\right]=0$

B-2, 4 $f'(x+y)=f(x).f'(y)$ (keeping x as a constant)
$\Rightarrow f'(x)=2f(x) \Rightarrow \log f(x)=2x+c$
$\Rightarrow f(x)=ce^{2x} \Rightarrow f(0)=1 \Rightarrow c=1$

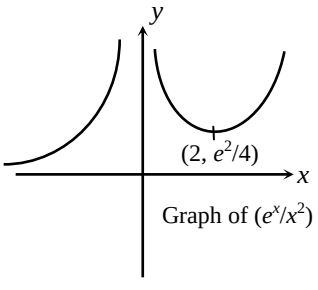
So $f(x)=e^{2x}$

$f\left(\frac{x}{2}\right)=e^{x}$

Now $\frac{e^{x}}{x^{2}}=k$

For $x>0$ minimum value of $\frac{e^{x}}{x^{2}}$ is $\frac{e^{2}}{4}$

$\Rightarrow$ For exactly one solution $0<k<\frac{e^{2}}{4}$.



Graph of (e^x/x^2)

C-2, 3, 4 $\int_{-2}^{-1} 1\,dx+\int_{-1}^{1} 0\,dx+\int_{1}^{2} 1\,dx=2$

D-1, 4 $g(x)=\sin^{-1}([x]+1)+\cos^{-1}[x]$, domain $-1\le x<1$

Range of $g(x)=\pi$

$f(x)+g(x)=3$

$\cos^{-1}(\cos x)=3-\pi$

This is not possible, hence no solution.

2. (A)-3, 4 Hence $(1+\tan k)(1+\tan(45-k))=(1+\tan k)\left(1+\frac{1-\tan k}{1+\tan k}\right)$

It follows that

$(1+\tan 1°)(1+\tan 2°)...(1+\tan 45°)$

$=(1+\tan 1°)(1+\tan 44°)(1+\tan 2°)(1+\tan 43°)$

$\ldots(1+\tan 22°)(1+\tan 23°)(1+\tan 45°)$

$=2^{23}$.

$\Rightarrow \lambda+1=24$

(B)-4 Solving $x+2y+3z=7,\ 2x-y+z=5$ we get $x-3y-2z=-2$

So if $\lambda = 2$, then

$x-3y-2z=-2$ and $x-3y-2z=1$ are in-consistent.

(C)-3, 5 Clearly the line is parallel to the plane.

If $(1,-|\lambda|,\ 0)$ lies on the plane $2x-y-z=5$

$2+|\lambda|-0=5 \Rightarrow \lambda=\pm 3$

(D)-1, 4 $A(1,\ 0,\ 0),\ B(0,\ a,\ 0),\ C(0,\ 0,\ a)$

Area of $\Delta ABC=\frac{1}{2}|a|.\sqrt{2+a^{2}}=\sqrt{6} \Rightarrow a=\pm 2$

SECTION–V
Integer Answer Type

1. Replace x by $-x$

$$2f(-x)+f(x)=\frac{1}{x}\sin\left(x-\frac{1}{x}\right)$$

and solving with the given equation, we get $3f(x)=\frac{1}{x}\sin\left(x-\frac{1}{x}\right)$

$$\Rightarrow I=\frac{1}{3}\int_{1/e}^{e}\frac{1}{x}\sin\left(x-\frac{1}{x}\right)dx \qquad \text{put } x=\frac{1}{t}$$

$$I=\frac{1}{3}\int_{e}^{1/e}t\sin\left(\frac{1}{t}-t\right)\left(-\frac{1}{t^2}\right)dt=-\frac{1}{3}\int_{1/e}^{e}\frac{1}{t}\sin\left(t-\frac{1}{t}\right)dt=-I$$

$$\Rightarrow I=0$$

Ans. 0

2. Probability of hitting the target in one fire $p=\frac{1}{5}$

Then probability of hitting the target at least once in n fires

$=1-$(probability of not hitting the target)

$$=1-\left(\frac{4}{5}\right)^n>\frac{3}{4}$$

$$\Rightarrow \left(\frac{4}{5}\right)^n<\frac{1}{4} \text{ as } \left(\frac{4}{5}\right)^6>\frac{1}{4} \text{ and } \left(\frac{4}{5}\right)^7<\frac{1}{4}$$

The least value of $n = 7$

Ans. 7

3. $\sum_{r=0}^{200}\alpha_r x^r=\sum_{r=0}^{200}\beta_r(1+x)^r$

$\alpha_0+\alpha_1x+\alpha_2x^2+.....+\alpha_{200}x^{200}$

$=\beta_0+\beta_1(1+x)+.....+\beta_{200}(1+x)^{200}$

Equating coefficient of x^{100}, we get $\alpha_{100}={}^{100}C_{100}+{}^{101}C_{100}+.....\ {}^{200}C_{100}={}^{201}C_{101}$

Similarly we can find $\alpha_{101},.....\alpha_{200}$.

$\sum_{r=100}^{200}\alpha_r={}^{201}C_{101}+{}^{201}C_{102}+.....\ {}^{201}C_{201}$

$A=2^{200}$

When A is divided by 15 remainder is 1.

Ans. 1

4. In the first round all integers, which leave remainder 1 when divided by 15, will be marked,

last number of this category is 991. Next number will be 991 + 15 – 1000 = 6. That means in second round all integers, which leave remainder 6 when divided by 15 will be marked. In short numbers of the form $5k + 1$ will be marked.

$\therefore \quad A = 200$ and $\frac{A}{50} = 4$.

Ans. 4

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